K-theory of Endomorphisms

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Suppose A is a ring and $End(A)$ is the category of pairs $(P, \alpha : P \to P)$, where P is a finitely projective A-module and α is A-linear. Let $Nil(A)$ denote the subcategory of $End(A)$ given by requiring α to be nilpotent.

Let $\tilde{K}End(A)$, $\tilde{K}Nil(A)$, and $\tilde{K}[x])$ be the homotopy fibers of the maps into $K(A)$ from $KEnd(A)$, $KNil(A)$, and $K[x])$ respectively.

Recall the *fundamental theorem of algebraic K-theory.*

- $\Omega\tilde{K}[x]) \simeq \tilde{K}(Nil(A)$, and
- $\Omega KA[x, x^{-1}] \simeq \Omega KA[x] \times KNil(A)$.

The goal of this talk is to explain an analogous result where $Nil(A)$ is replaced with $End(A)$. This result uses Dwyer's notion of *derived* Cohen localizations of rings and the associated K-theory fibrations sequence. One motivation for this work is the study of families of endomorphisms of topological spaces.